

Does forcing JSON make small local LLMs worse?

I ran a controlled experiment on my laptop to find out.

Plus: the bug that made my benchmark lie about speed.

Swipe →

THE SETUP

Only the format changes

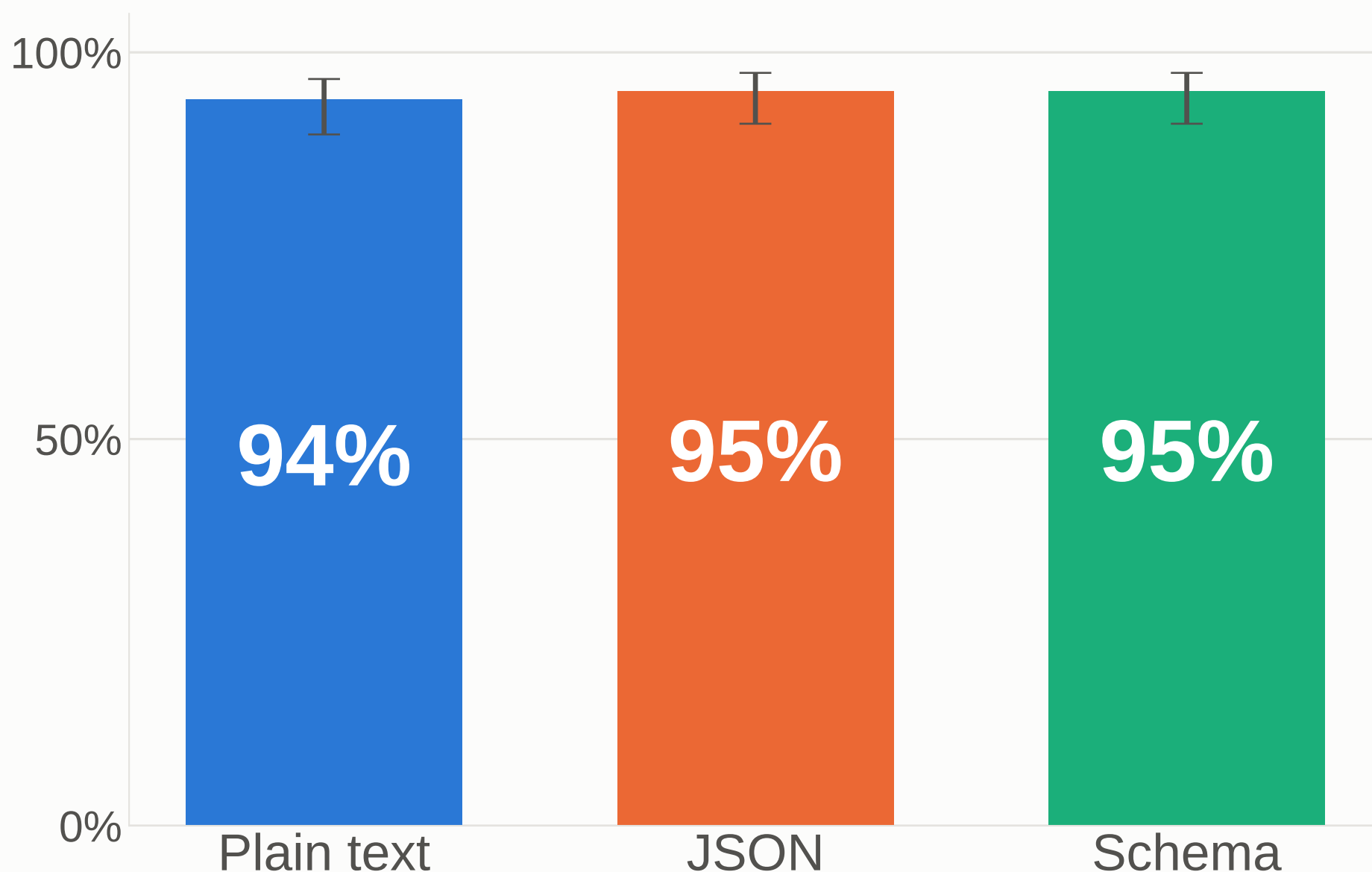
3 models × 3 output modes × 60 questions

-  **Plain text**
labelled lines, no constraint
-  **JSON mode**
valid JSON enforced
-  **Schema**
decoding locked to the schema

Same instructions, same context,
one validation contract (Pydantic).
Phi-3 3.8B · Mistral 7B · Gemma 2 9B
via Ollama on an RTX 4070 laptop.

FINDING 1

JSON cost no measurable accuracy

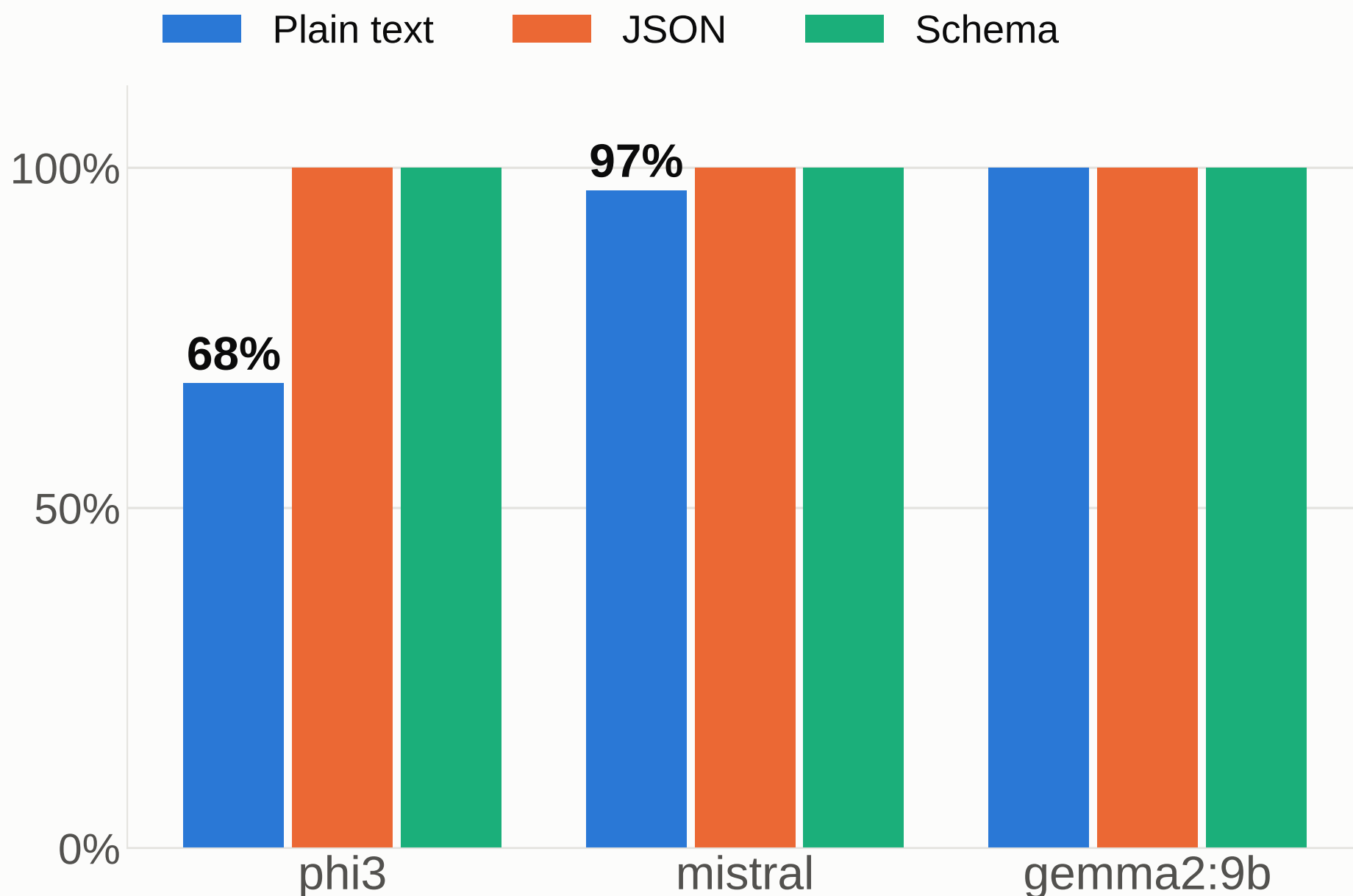


JSON vs text: +1.1 pp (95% CI -1.1 to +3.3)

Paired test on the same questions.

FINDING 2

But it made output machine-readable



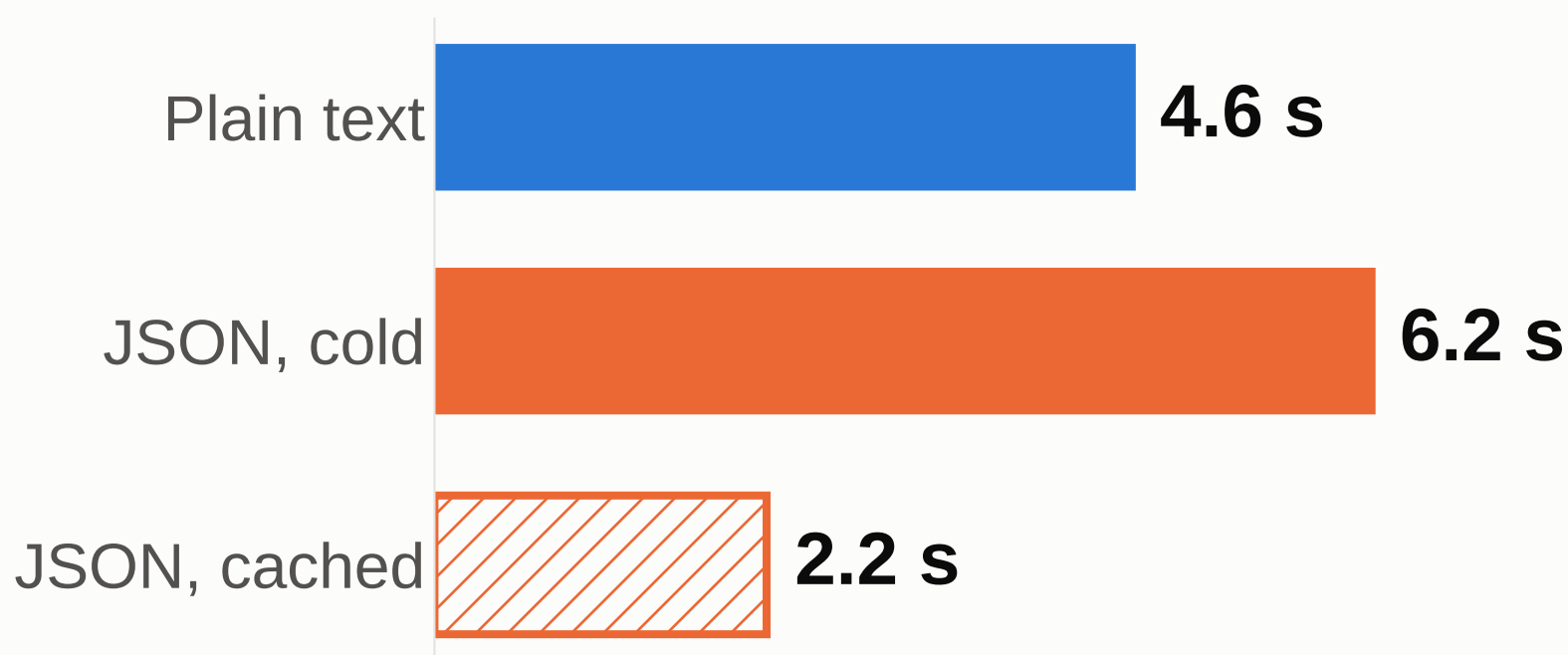
Valid on first try: 88% (text) → 100% (JSON)

Phi-3 kept dropping the 'Answer:' label.

FINDING 3 · THE TRAP

My benchmark said JSON was faster. It wasn't.

gemma2:9b, median response time



JSON and schema prompts were identical, so Ollama's prompt cache gave the second one a free head start.

Cold-only: 0.93–1.34× plain text.
Fixed with a per-request nonce.

FINDING 4

**"100% confident"
barely means
anything**

475/540

answers stated confidence = 1.0

- 18 were wrong at ≥ 0.9 confidence
- Phi-3: "33 days" ($28 + 2 = 30$),
at 1.0, in all three modes
- AUROC 0.58: barely better
than a coin flip (0.5)

**Don't route answers to humans
on self-reported confidence.**

TAKEAWAYS

What I'd tell an AI team

- 1 Use structured output.**
For grounded Q&A it cost no measurable accuracy.
- 2 Benchmark latency cold.**
Shared prompt prefixes fake speed-ups through the cache.
- 3 Don't trust stated confidence.**
Check retrieval or use a verifier instead.
- 4 Report your parser.**
'Reliability' gains depend on how strict it is.

How it's built

- Python · Ollama · Pydantic
- Paired bootstrap CIs + McNemar tests
- Calibration: ECE, Brier, AUROC
- No-egress privacy guard, tested in CI
- 46 tests · interactive dashboard
- Findings generated from the data

Results, code and all answers:

abdulmuhaminkhan.github.io/privacy-first-llm-eval

(link in my Featured section)

**Next: newer 2026 models
and quantisation.**